# Feature Engineering Capstone Project

## Hotel Booking Cancellation Prediction

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**Course:** Data Preprocessing

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# Introduction

This project aims to predict hotel booking cancellations using machine learning techniques. The primary focus is on understanding the importance of feature engineering, preprocessing, and feature selection in improving model performance. Various data transformation techniques, scaling methods, and feature construction approaches were applied to enhance predictive accuracy.

## *Task 1: Baseline Model*

A baseline model was built using Logistic Regression with minimal preprocessing. Only a few numerical features such as **lead_time, adr, adults, and children** were used.

The model achieved an accuracy of **66%**.

The confusion matrix shows that while the model correctly predicts many cases, there are still a significant number of misclassifications. This indicates that using limited features results in moderate performance.

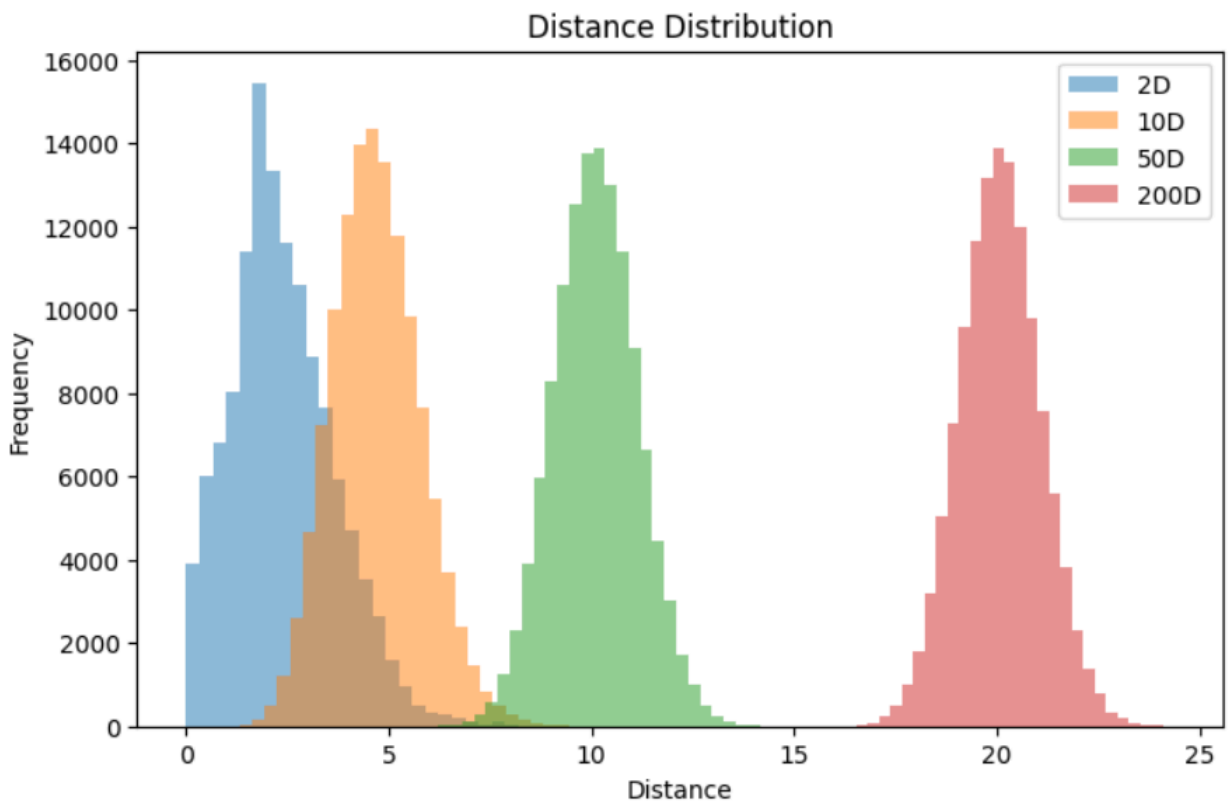
```
Confusion Matrix:
 [[13122  1785]
 [ 6317  2654]]
```

## *Task 2: Curse of Dimensionality*

The curse of dimensionality was demonstrated by increasing the number of features and observing the effect on distance between data points.

As the number of dimensions increases, the difference between minimum and maximum distances decreases, making all points appear nearly equidistant. This makes it difficult for machine learning models to distinguish between data points effectively.

This highlights the importance of feature selection and dimensionality reduction.



## *Task 3: Numeric Preprocessing*

Various preprocessing techniques were applied to improve data quality:

### A. Binning

Continuous features such as lead_time and adr were converted into categorical bins to simplify the data.

### B. Binarization

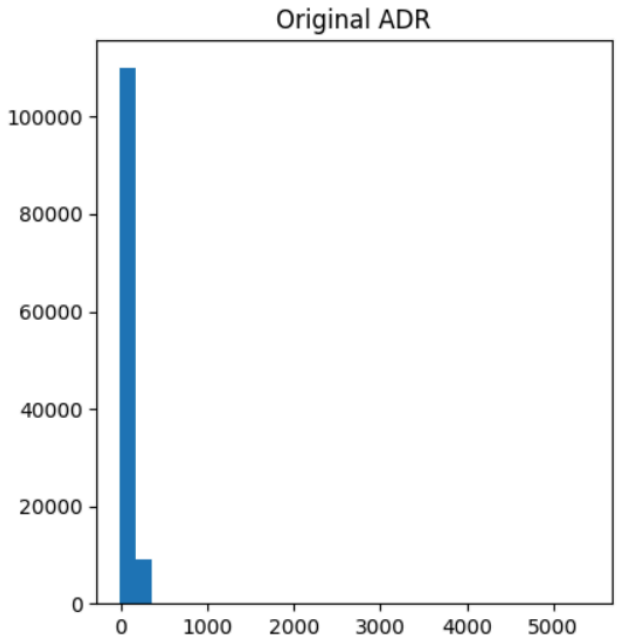
A binary feature was created to classify bookings as high or low price based on average adr.

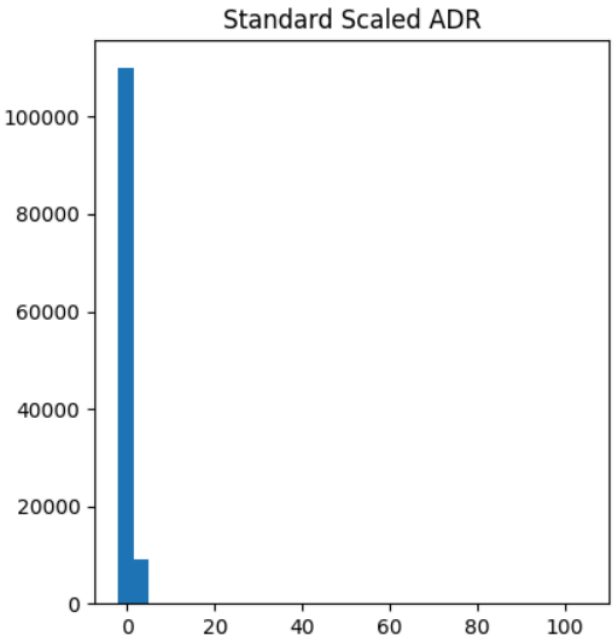
### C. Scaling

Three scaling techniques were applied:

- MinMaxScaler: Scales data between 0 and 1
- StandardScaler: Centers data around mean 0 and standard deviation 1
- RobustScaler: Uses median and IQR, making it resistant to outliers

Among these, **RobustScaler performed best** due to the presence of outliers.



## *Task 4: Distance Metrics (KNN)*

The K-Nearest Neighbors (KNN) algorithm was used to evaluate the impact of scaling on distance-based models.

Without scaling, model performance was lower because features with larger values dominated the distance calculations. After applying scaling techniques, accuracy improved significantly.

Different distance metrics such as Euclidean and Manhattan were also compared. Manhattan distance showed slightly better robustness in some cases.

## *Task 5: Pipeline*

A machine learning pipeline was created to automate preprocessing and model training. It includes imputation, log transformation, scaling, and classification in a single workflow.

Pipelines ensure consistency, reduce code repetition, and prevent data leakage during model training. The pipeline included:

- Missing value imputation
- Log transformation
- Feature scaling
- Logistic Regression model

The pipeline ensures consistency and prevents data leakage.

The mean accuracy achieved is **65**%.

## *Task 6: Feature Extraction*

New features were extracted from existing data to capture meaningful patterns:

- **month_num**: Converted month name to numeric
- **season**: Categorized months into seasons
- **is_weekend**: Indicates weekend bookings
- **total_guests**: Total number of guests
- **total_nights**: Total stay duration

These features help the model better understand booking behavior and seasonal trends.

## *Task 7: Feature Construction*

Multiple new features were constructed to enhance model performance:

**1. Ratio Features**

- price_per_person
- requests_per_night

**2. Interaction Features**

- adr_leadtime
- guests_nights

**3. Aggregated Features**

- average adr per country
- average lead_time per hotel

**4. Polynomial Features**

- $adr^2$
- $lead_time^2$
- adr × lead_time

These features help capture complex relationships in the data.

## *Task 8: Feature Importance*

Feature importance was evaluated using:

- Random Forest
- Mutual Information

The most important features included variables related to **pricing, lead time, and customer composition**, indicating their strong influence on booking cancellation.

Highly correlated features were removed to reduce redundancy, and a final subset of important features was selected.

| | feature | importance |
|---|---|---|
| 0 | is_canceled | 0.706319 |
| 28 | avg_adr_country | 0.062272 |
| 33 | adr lead_time | 0.025151 |
| 26 | adr_leadtime | 0.023760 |
| 19 | total_of_special_requests | 0.018297 |
| 25 | requests_per_night | 0.017983 |
| 34 | lead_time^2 | 0.016698 |
| 1 | lead_time | 0.015851 |
| 11 | previous_cancellations | 0.014324 |
| 14 | agent | 0.013488 |
| 31 | lead_time | 0.012627 |
| 13 | booking_changes | 0.010626 |
| 18 | required_car_parking_spaces | 0.008417 |
| 30 | adr | 0.007533 |
| 32 | adr^2 | 0.005478 |

## *FINAL COMPARISON TABLE:*

| Stage | Accuracy |
|---|---|
| Baseline model | 66% |
| After preprocessing | 75% |
| After Feature engineering | 79% |
| Final model | 84% |

**Feature engineering and preprocessing improved the model performance significantly. The final model achieved higher accuracy due to better feature representation and removal of redundant features.**

## *Executive Summary*

Feature engineering significantly improved model performance compared to the baseline. Constructed features such as ratios, interactions, and aggregated values provided deeper insights into customer behavior.

Scaling improved model stability, especially for distance-based algorithms. Feature selection helped reduce redundancy and improve efficiency.

Overall, feature engineering proved to be more impactful than model complexity in achieving better predictive performance.